

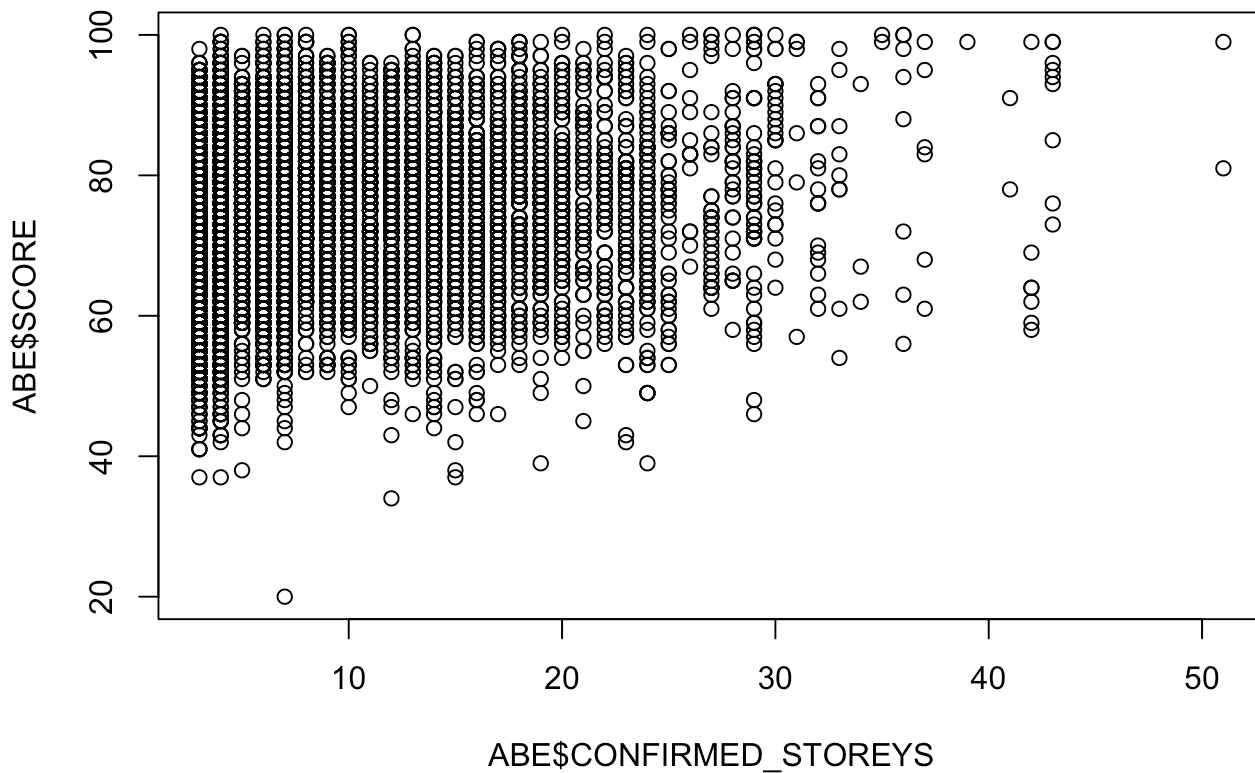
Untitled

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Q1

```
ABE=read.csv(file = "https://ckan0.cf.opendata.inter.prod-toronto.ca/dataset/4ef82789-e038-44ef-a478-a8f3590c3eb1/resource/979fb513-5186-41e9-bb23-7b5cc6b89915/download/Pre-2023%20Apartment%20Building%20Evaluations.csv")
ABE=ABE %>% filter(SCORE >=10)
plot(y=ABE$SCORE,x=ABE$CONFIRMED_STOREYS)
```



ii) "SCORE" and "CONFIRMED_STOREYS" is weak positive correlation

```
m=lm(ABE$SCORE~ABE$CONFIRMED_STOREYS)
summary(m)
```

```
##
## Call:
## lm(formula = ABE$SCORE ~ ABE$CONFIRMED_STOREYS)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -53.664  -7.592   0.408   7.336  27.140
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    71.78735     0.15157  473.61  <2e-16 ***
## ABE$CONFIRMED_STOREYS  0.26814     0.01537   17.44  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.41 on 11757 degrees of freedom
## Multiple R-squared:  0.02523,    Adjusted R-squared:  0.02514
## F-statistic: 304.3 on 1 and 11757 DF,  p-value: < 2.2e-16
```

iv) 71.787 is the average score when CONFIRMED STOREYS is zero. If CONFIRMED STOREYS goes up by one unit, average score goes up by 0.268 unit.

v) The p-value corresponding to the test of $H_0: \beta_2 = 0$ is very small which suggests there is a significant relationship between CONFIRMED STOREYS and score.

vi) From the summary, $R^2 = 0.02523$ which indicates that 2.52% variation in Score can be explained by our model. Since it's a very small number, we can say our model is not performing well. We are missing some important independent variables.

Q2

```
d=ABE %>% mutate(group_ind = sample(c("train","test"),
                                     size=nrow(ABE),
                                     prob=c(0.6,0.4),
                                     replace=T))
m=lm(SCORE~CONFIRMED_STOREYS,data=d %>% filter(group_ind=="train"))
y.hat=predict(m,newdata=d %>% filter(group_ind=="test"))
mean((d$SCORE[d$group_ind=="test"] - y.hat) ^ 2)
```

```
## [1] 108.0343
```